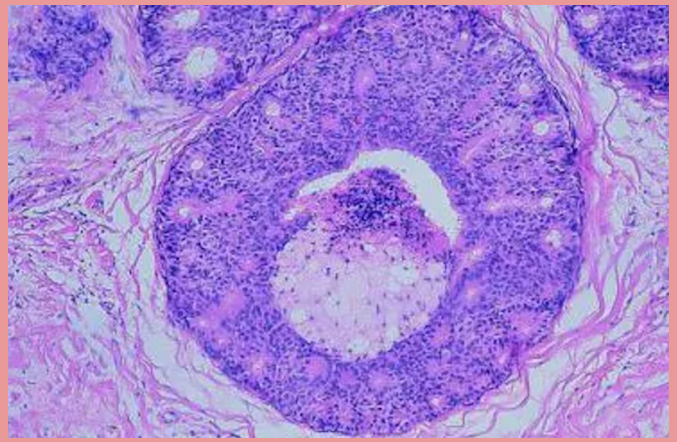
Predicting breast cancer

LINH, LEAH, JULES, ERIC, KANNASICHO

RESEARCH QUESTION:

What tumor features are the most predictive of malignancy in breast cancer using US data

Background on Breast Cancer



1. Prevalence and Detection

Breast cancer is the most commonly diagnosed life-threatening cancer in women worldwide and is often detected early through screening mammography before symptoms appear.

2. Diagnostic Approach

Standard evaluation includes clinical breast examination, imaging (mammography / ultrasound), and needle biopsy, which together improve early diagnosis and treatment planning.

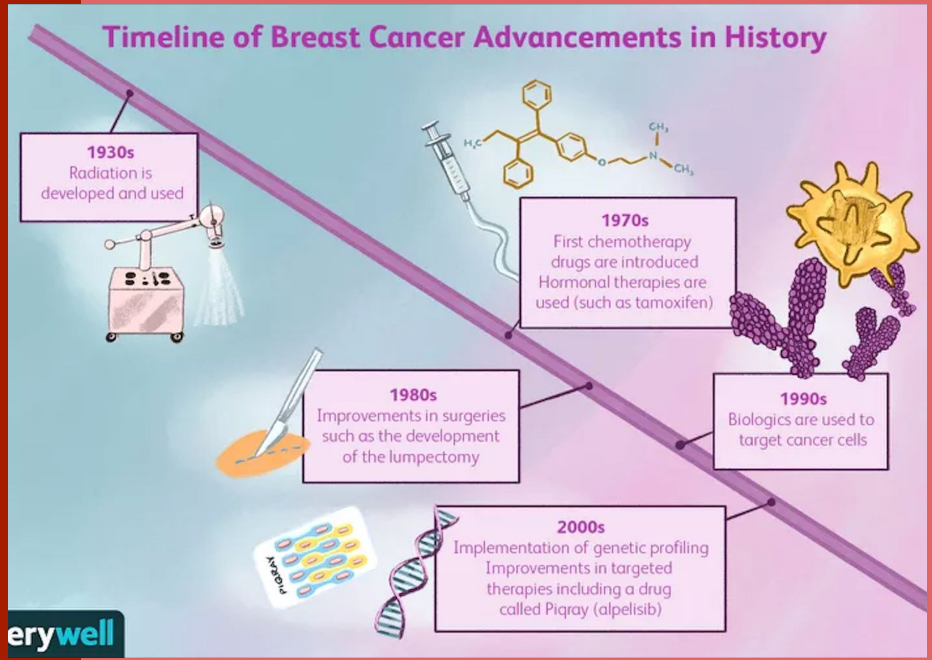
3. Treatment and Outcomes

Primary treatment involves surgery and radiation, with hormone or chemotherapy added when needed. Early-stage, low-risk cases are often curable, and adjuvant therapies significantly reduce recurrence.

4. Advances in Research

Over the past 30 years, major research efforts have deepened understanding of breast cancer biology and led to more effective, targeted, and less toxic treatment options.

History of Breast Cancer



1930s – Radiation Therapy

- Radiation developed and used as one of the earliest non-surgical treatments
- Provided an alternative to radical mastectomy
- Improved local tumor control

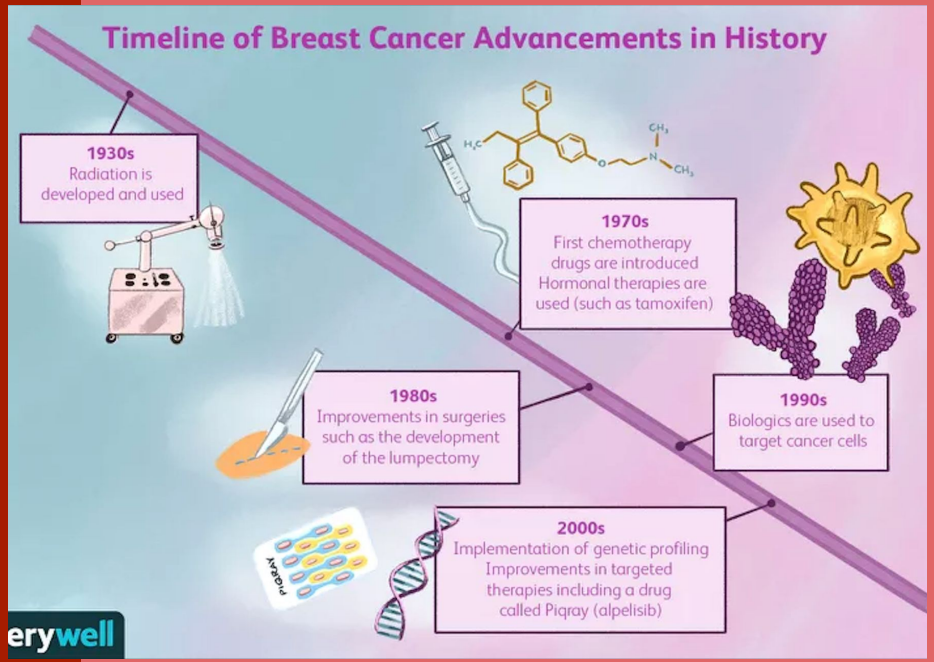
1970s – Chemotherapy & Hormonal Therapy

- First chemotherapy regimens introduced (e.g., CMF)
- Tamoxifen becomes the first widely used hormonal therapy
- Beginning of biology-based treatment decisions

1980s – Advances in Surgery

- Shift to breast-conserving surgery (lumpectomy + radiation)
- Equivalent survival to mastectomy for early-stage cancer
- Early development of sentinel lymph node biopsy

History of Breast Cancer



1990s – Biologic Therapies

- Rise of biologic therapies targeting specific cancer pathways
- HER2-targeted therapy (trastuzumab/Herceptin) revolutionizes care
- Improved outcomes for aggressive subtypes

2000s – Genetic Profiling & Targeted Therapies

- Genetic tests (Oncotype DX, MammaPrint) guide treatment decisions
- Targeted therapies expand (e.g., PI3K inhibitors, PARP inhibitors)
- Move toward precision oncology and personalized treatment

Datasets description

- Cleaned data set, no null value
- 31 variables, 569 samples
- Each sample represents a patient case where the tumor has already been medically evaluated
- from ucimlrepo import fetch_ucirepo

```
Target value counts:
target
1    357
0    212
```

!

Potential Variables

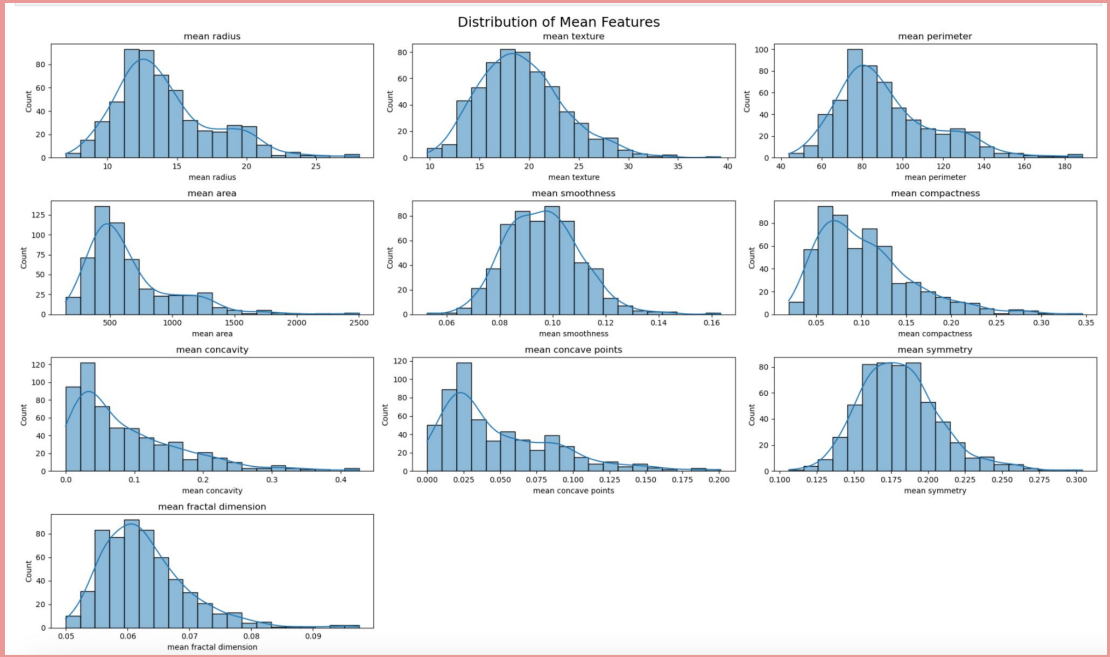
3 main variable group:

- Mean (average size/shape/texture/...): mainly descriptive data

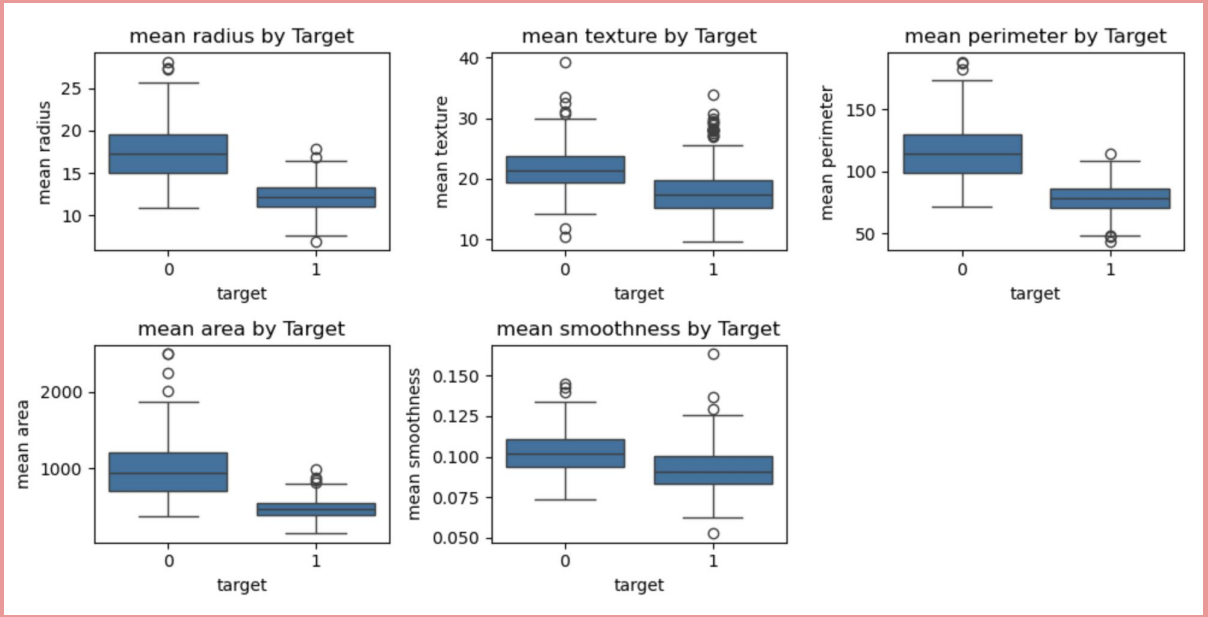
- Worst (most abnormal recorded): correlation for predictive modeling

- Target (0 = Malignant (cancerous) / 1 = Benign (non-cancerous)): learning patterns that distinguish cancerous vs benign tumors from past cases.

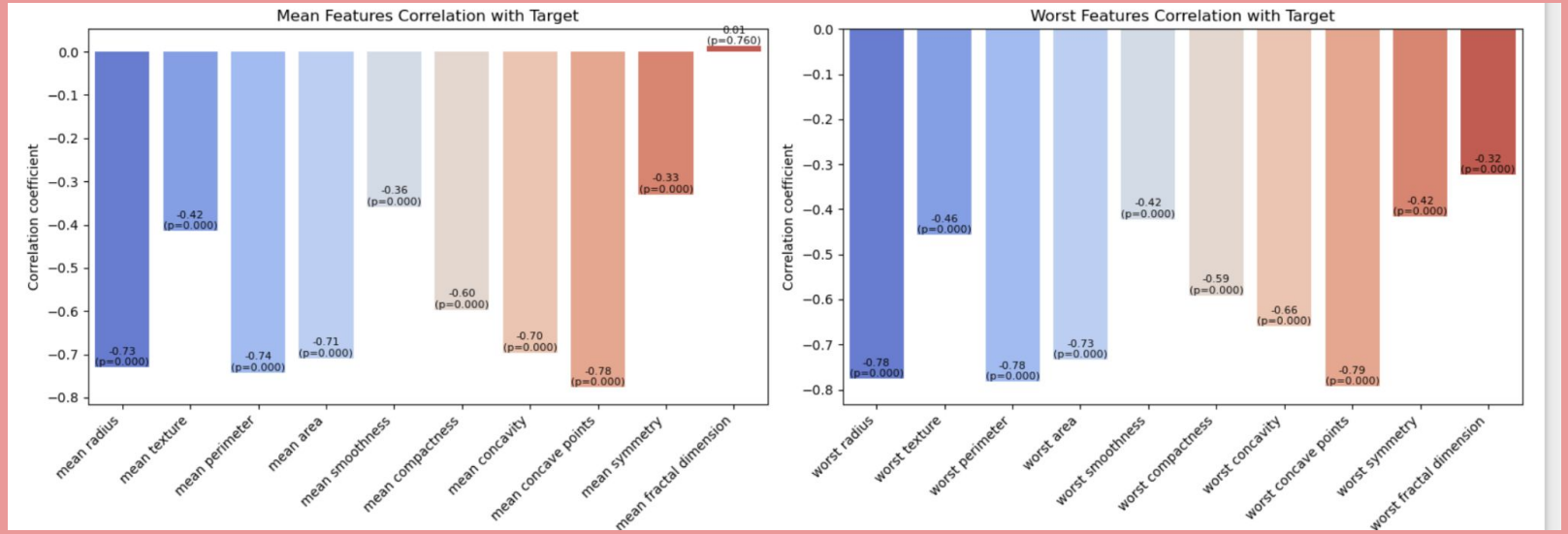
Descriptive Visualization



Descriptive Visualization



Correlation: Worst/Mean group vs Binary Target

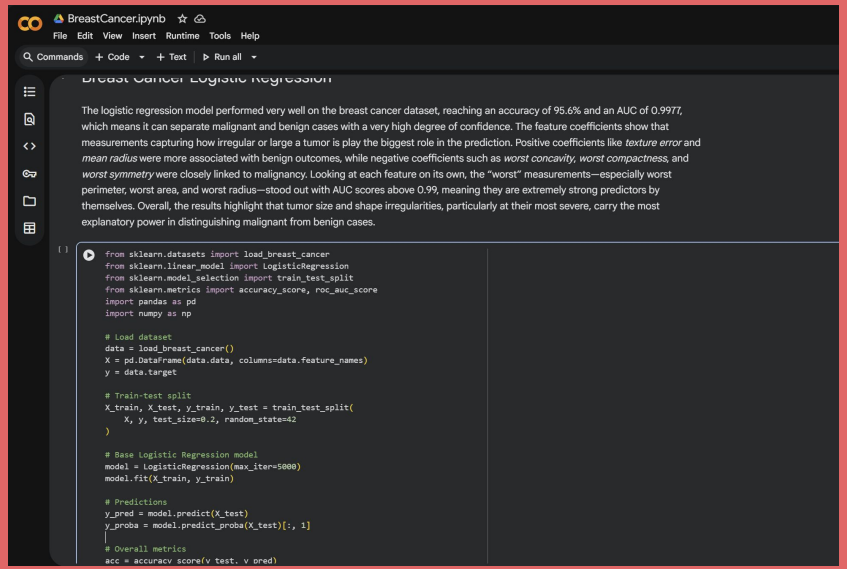


- Negative correlation → feature increases → tumor more likely malignant.
- Positive correlation → feature increases → tumor more likely benign.

MODELING

- Logistic regression
- K-Nearest Neighbors
- Random Forest

Logistic regression



- Accuracy of 95.6% and an AUC of 0.9977
- Irregularity or largeness of a tumor is play the biggest role in the prediction
- Coefficients like texture error and mean radius were more associated with benign outcomes
- Worst concavity, worst compactness, and worst symmetry were closely linked to malignancy

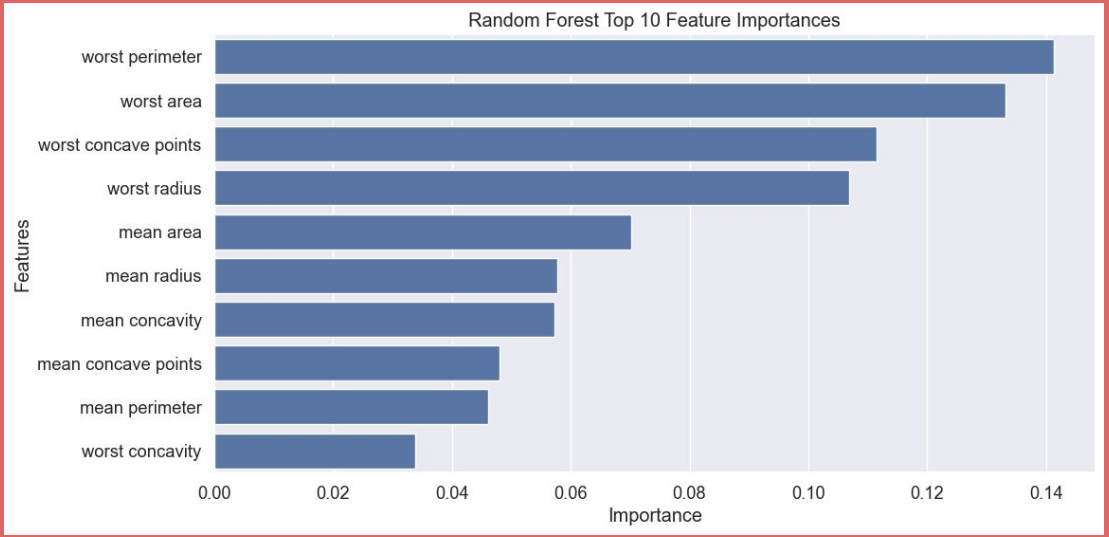
KNN + Random Forest

Most defining features:

- Worst perimeter, area, concavity, radius
- Mean area, radius, concavity, perimeter

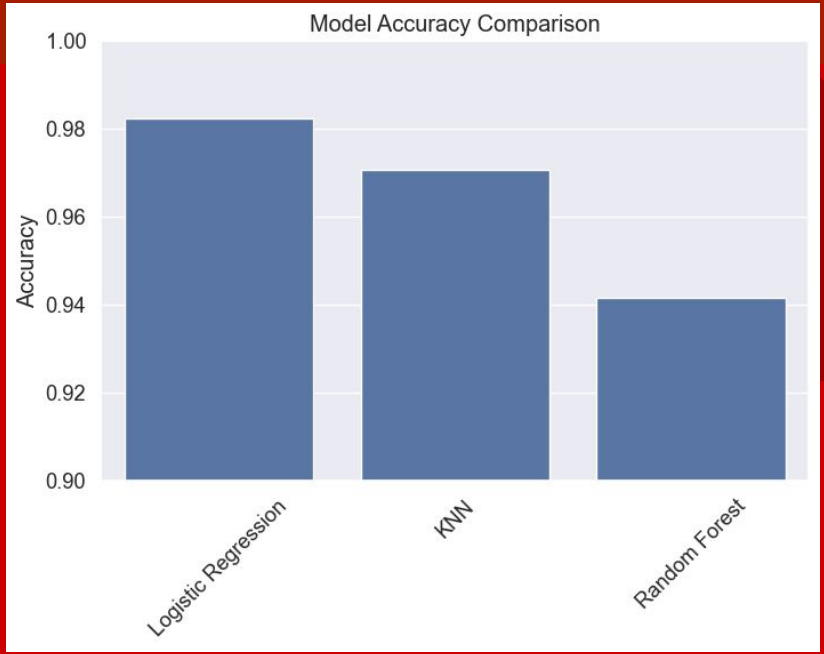
KNN Accuracy: 97.1%

RF Accuracy: 94.2%



MODEL RESULTS

- All models perform pretty well
- Logistic Regression has the highest accuracy
- Random Forest captures non linear relationships

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- Logistic Regression -
Accuracy: 0.9825
Confusion Matrix:
[[ 62   2]
 [  1 106]]
```

```
- KNN -
Accuracy: 0.9708
Confusion Matrix:
[[ 59   5]
 [  0 107]]
```

```
- Random Forest -
Accuracy: 0.9415
Confusion Matrix:
[[ 58   6]
 [  4 103]]
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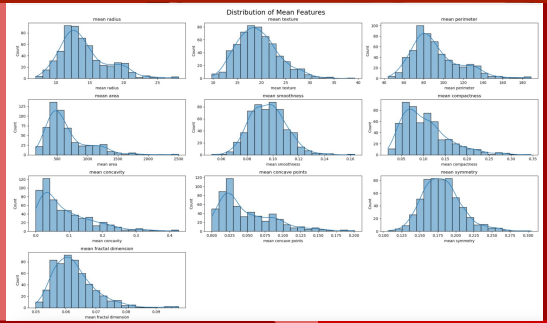
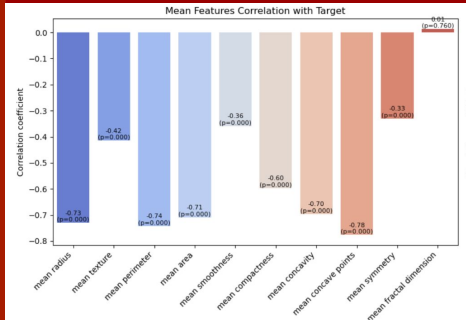
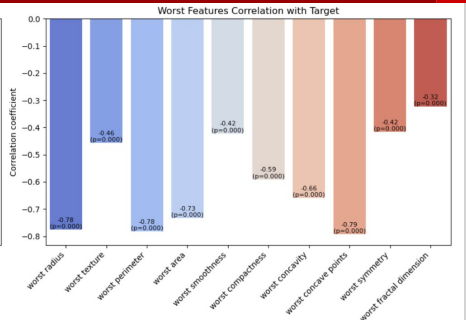
RESULTS



- Most mean features (size, texture, symmetry) show right-skewed distributions, indicating a wide range of tumor variability.
- Benign vs malignant tumors visually differ in spread and clustering across features

- "Worst" measurements (most abnormal observation for each cell nucleus) consistently had stronger correlations than mean features.
- Indicates that the most extreme tumor abnormalities are the most predictive of malignancy.





Strong negative correlations with the malignant class (0 = malignant, 1 = benign): Worst radius, worst perimeter, worst area, mean radius, mean concavity, mean texture *(Higher values → more likely malignant)*
Weak positive correlations appear in a few features, slightly associated with benign tumors.

Why did we choose to address breast cancer?

SPOILER: WE LOVE WOMEN

Conclusion

- Tumors with larger size and more irregular borders show a strong association with malignancy.
- “Worst” values capture abnormalities more effectively than average values → important for early diagnostic models.
- Simple imaging-derived features (like radius, area, concavity) already provide strong predictive signals without needing clinical data.
- Findings support the hypothesis that tumor morphology is a powerful indicator of malignancy.
- These results highlight the potential of using quantitative tumor features to improve screening accuracy and guide clinical decision-making.

Limitations

Dataset & Feature Limitations

- Dataset includes only morphological features—no clinical factors (age, race, ER/PR, HER2).
- Cannot capture biological behavior or subtype differences (e.g., triple-negative vs HER2+).

Generalizability Issues

- Dataset is relatively small (569 samples) and collected from a U.S. setting → possible population bias.
- May not generalize well to global or diverse patient groups.

Methodological Constraints

- Correlation does not imply causation—results show association, not biological mechanism.
- Many features are mathematically related (radius ↔ perimeter ↔ area), leading to redundancy.
- Imaging features alone cannot replace comprehensive diagnostic evaluation.

References

1. Pavani Chalasani, M. (2025, October 10). *Breast cancer clinical presentation*. History, Physical Examination.

2. Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5DW2B.